```
#include <stdio.h>
#include <ctype.h>

#define MAX 100

int stack[MAX];
int top = -1;

// Push operation
void push(int num)
{
  top++;
  stack[top] = num;
}

// Pop operation
int pop()
{
  int num;

  num = stack[top];
  top--;

  return num;
}

// Postfix evaluation
int evaluatePostfix(char postfix[])
{
  int i, a, b, result;
  char ch;

  top = -1;

  for (i = 0; postfix[i] != '\0'; i++)
  {
    ch = postfix[i];

    // If operand
    if (isdigit(ch))
    {
      push(ch - '0');
    }

    // If operator
```

```
      else
      {
         b = pop();
         a = pop();

         switch (ch)
         {
            case '+':
               result = a + b;
               break;

            case '-':
               result = a - b;
               break;

            case '*':
               result = a * b;
               break;

            case '/':
               result = a / b;
               break;

            case '^':
            {
               int j;
               result = 1;

               for (j = 0; j < b; j++)
                  result = result * a;

               break;
            }
         }

         push(result);
      }
   }

   return pop();
}

// Main function
int main()
{
```

```
    char postfix[MAX];
    int result;

    printf("Enter postfix expression: ");
    scanf("%99s", postfix);

    result = evaluatePostfix(postfix);

    printf("Result = %d\n", result);

    return 0;
}
```